

13	A sealed container is filled with an ideal gas which is initially at a temperature of 50 °C. The root-mean-square (r.m.s.) speed of the gas molecules at this temperature is measured to be 400 m s⁻¹. If the temperature of the gas is increased to 100 °C, what is now the r.m.s speed of the gas molecules?
	A 430 m s⁻¹ B 460 m s⁻¹ C 570 m s⁻¹ D 800 m s⁻¹
	Answer: A $\frac{1}{2}m(c^2) = \frac{3}{2}kT$ $c_{rms} \propto \sqrt{T}$ $\frac{c_{rms,new}}{400} = \sqrt{\frac{100+273.15}{50+273.15}} = 1.07$ $c_{rms,new} = 430 \text{ m s}^{-1}$

[Turn over

14	Which of the following is not an example of resonance?
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